

NIKHIL DHILLON

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EDUCATION

University of Victoria

Bachelor of Science in Honours Computer Science

GPA: 8.75 / 9.0

Victoria, BC

Expected December 2026

SKILLS

- **Programming Languages:** Python, Java, C, C++, HTML, CSS, JavaScript, AVR Assembly, SQL (PostgreSQL), R
- **Frameworks and Libraries:** Django, PyTorch, Spring Boot, Express.js, React.js, React Native, Node.js, Flask, JUnit, Selenium, BeautifulSoup4, Playwright, Pandas, NumPy, Matplotlib
- **DevOps and Tools:** Git, Docker, Microsoft Azure, Google Cloud Platform, Linux, Bash, DBeaver, FileZilla, VS Code, PyCharm, IntelliJ IDEA, REST API, Bruno
- **Project Management and Collaboration:** Slack, Microsoft Teams, Jira, Confluence, Scrum, Agile

PROFESSIONAL EXPERIENCE

Reliable Controls

Software QA Analyst

Victoria, BC

September 2024 - April 2025

- Architected and optimized test case workflows in TestRail, validating over 100 test cases for functional and non-functional requirements, and ensuring compliance with QA standards and testing best practices.
- Streamlined API testing and integration validation workflows with Bruno, increasing reliability and adherence to API specifications, while reducing API validation errors by 20%.
- Spearheaded the successful delivery of the Offline Engineering Configuration project for high-profile clients, meeting tight deadlines and ensuring deployment of critical features without post-deployment issues.
- Reduced the average QA review cycle time through process optimization and enhanced automation strategies, as reflected in KPI metrics, led to increased sprint velocity.

PERSONAL PROJECTS

Wearlyze | Pytorch, Django, Deeplabv3+, EfficientNetV2, REST API

October 2024 - Present

- Architected and developed Wearlyze, an AI-driven outfit recommendation platform that identifies clothing items in user-uploaded images and retrieves matching retail products from an ever-updating database of over 500,000 items, revolutionizing personalized online shopping experiences.
- Enhanced and fine-tuned DeepLabV3+ with a ResNet-101 backbone on the DeepFashion2 dataset, comprising over 800,000 labeled clothing items, achieving over 85% mean Intersection over Union (mIoU) in accurately detecting and categorizing apparel items in user-submitted images.
- Leveraged PyTorch and integrated EfficientNetV2 for feature extraction, optimizing the recommendation engine to reduce inference time by 40% and achieving a 90% match rate for retail product recommendations.

Returnly | NodeJS, ExpressJS, React Native, Sequelize, PostgreSQL

January 2025 - Present

- Developed a full-stack parcel return platform, reducing expected return processing times by 40% through a React Native frontend and a Node.js backend.
- Optimized API performance, reducing response times by 30% through query restructuring, load balancing with Nginx, and Redis caching.
- Implemented secure user authentication with Passport.js, reducing authentication failures by 50% through JWT-based session management and OAuth integration.
- Engineered real-time delivery driver tracking, enhancing transparency and improving drop-off time accuracy through GPS integration, live location sharing, and route optimization algorithms.

Budget Buddy | Python, Selenium, SMTP, BeautifulSoup4

August 2023

- Devised an automated web scraping application in Python operating the BeautifulSoup library to extract product information from Amazon, demonstrating proficiency in web scraping and data extraction techniques.
- Continuously monitors product details and prices on Amazon.ca at user-defined intervals (e.g., every 30 minutes).
- Simplified price tracking by sending automated email alerts (up to 15 weekly) whenever products meet your budget, allowing you to choose when to buy.